

NetSI PhD Bootcamp 2026 · Thursday, Sep 3 · 2pm–4pm

Agentic Computing

Coding agents in your PhD workflow

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How this session works

- ~45 min guided speed-run: what agents are, setup, the loop, tips
- ~60 min **Hackathon** pick one of three projects:
 - **Dashboard website · Extend your Session 1 work · Recreate a paper**
- ~15 min share-outs

Part 1: Speed-run

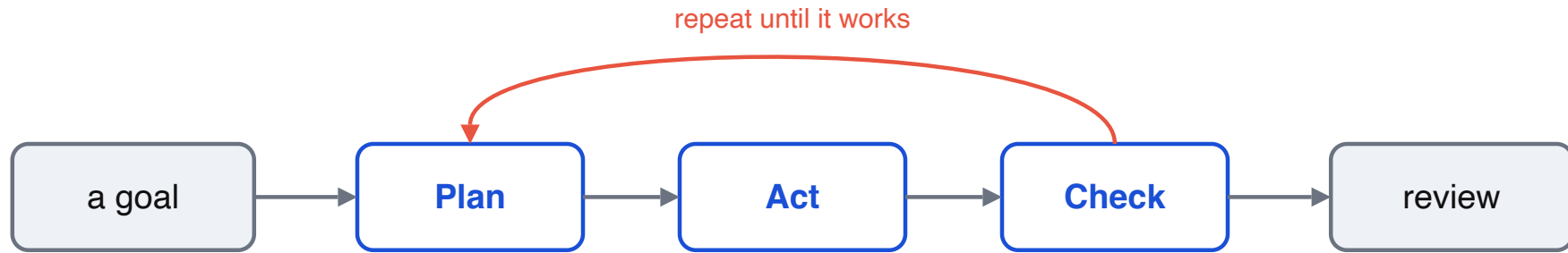
What is a coding agent?

An LLM that can read files, run commands, and edit code, in a loop

- You state a **goal**, it plans, acts, checks its own work, and repeats
- It works on your **whole project**, in your **terminal**, not just one snippet
- You stay the reviewer: it drafts, **you** decide what to keep

Think pair-programmer that can also run the tests and read the errors.

The loop



You give a goal. It plans, acts, and checks. You review the result.

Setup (do this now)

Hands-on Session

1. Go to claude.northeastern.edu, click **First time? Start here**, accept the guidelines
2. Log in with **SSO** (your NU email + credentials)
3. We'll use **Claude Code** (in the terminal) and **Claude Desktop**

Generous NU usage limits. Prefer something else? OpenAI Codex or opencode (open source) work too.

Claude Code in 30 seconds

Open a terminal **inside your project** and run `claude`:

```
$ claude
> load edges.csv and print the degree distribution

I'll write a short script, run it, and show you the output.
... (creates degree.py, runs it) ...
Top degrees: 42, 39, 31, ... Saved a histogram to degree.png.
```

It reads your files, proposes changes, and can run them. You approve as it goes.

Run it in **VS Code's integrated terminal** and it connects to the editor: reads your lint/type errors and shows edits as **inline diffs**.

The core loop in practice

- Ask for an **outcome**, not keystrokes: **"make the plot log-scale and rerun it"**
- Let it **plan first** on anything non-trivial, and read the plan before you approve
- Work in **small steps**: run it, check it, then continue
- When a task is done, **start fresh** so old context doesn't leak in

Best practice: a project memory file

`CLAUDE.md` tells the agent your conventions, it reads this file automatically

- Put it at the repo root. Claude Code loads it every session
- `/init` will draft one for you from the codebase
- Keep it short: stack, how to run things, do's and don'ts
- Prefer `AGENTS.md` (a cross-tool convention)? Add a line `@AGENTS.md` in `CLAUDE.md` so Claude reads it too

`CLAUDE.md`

- Python 3.11, networkx + matplotlib. Run jobs with ``sbatch``, not on the login node.
- Small diffs. Show me a plan before editing more than one file.

Tips: work with it, not against it

- **Plan mode**, `Shift+Tab` cycles into it. It reads the code and writes a plan before touching files
- **Make it prove it**, the #1 habit: a quick test, a linter, or a UI screenshot so it checks its own work
- **Keep context clean**, one task per session. `/clear` when you switch, `/compact` a long one
- **Be specific**, say what and where, so it doesn't read the whole project to guess

Git is your safety net

Commit before a big change, so you can always go back

- **Commit first**, before you let it make a big change
- `git diff` to review exactly what it changed
- Keep it? `git commit`. Want to toss one file? `git restore <file>` (discards just that file)
- Let the agent run git for you, but understand each step. That's what makes experimenting safe

Tips: power features (when you're ready)

- **Model choice**, use a stronger model to plan, a faster one to implement
- **Skills**, reusable `SKILL.md` guides that package a workflow or your house rules
- **Subagents**, spin up a helper for a side task (review, debug). It burns your limits faster
- **MCP**, connectors to external tools and data. Add sparingly to save context

Guardrails (say them out loud)

- **Read what it runs.** You own every command it executes
- **Verify claims.** "**Tests pass**" means **you** saw them pass
- **No secrets** in prompts (keys, passwords, private data)
- The agent is **confident even when wrong**. Stay skeptical

Read papers and find the literature

- **Read a paper:** pull out **claim** ▶ **evidence** ▶ **interpretation**, explain the method, check the math, find the weak points
- **At the start of a topic:** map the area, name the seminal papers, get a sensible reading order
- **Deep in it:** find the paper you're missing, what cites this, the related work you skipped
- **Verify:** agents **invent** citations. Confirm every reference actually exists

Synthesize and hypothesize

- **Synthesize:** combine findings across papers into a comparison table, surface contradictions and gaps
- **Generate hypotheses:** brainstorm mechanisms, ask "**what would falsify this?**", stress-test an idea before you sink weeks into it
- It widens the search. **You** decide which threads are worth chasing

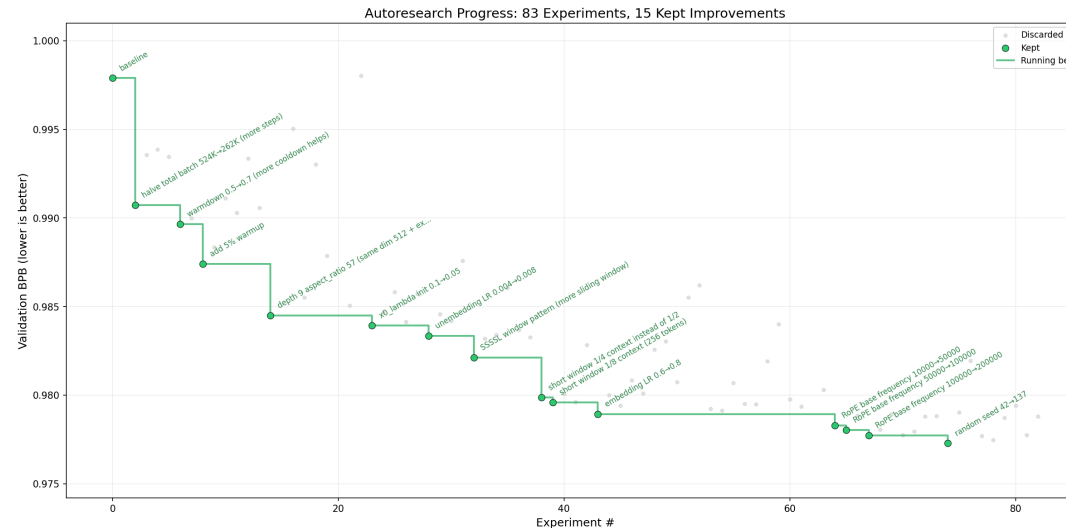
Prototype and unblock

- **Quick prototype:** turn an idea into a runnable script or notebook in minutes, just to sanity-check it
- **Escape dependency hell:** it reads the error, fixes the env, pins versions, gets you running again
- **The grunt work:** Slurm/conda, messy notebook ▶ clean script, reshape data, explain a stack trace, review for bugs

It drafts and scaffolds. **You** write the science and verify every result.

The frontier: auto research

Set a goal and guardrails, then let it loop overnight



Karpathy's [autoresearch](#): an agent edits `train.py`, runs ~100 five-minute experiments overnight, keeping the ones that lower validation loss (`val_bpb`).

Reality check: it drifts, over-claims, and burns compute. A tireless **assistant you supervise**, not a scientist. Verify everything.

Part 2: Hackathon

Three paths. Pick yours.

Choose your project

Track	Difficulty	You'll build
Dashboard website	BEGINNER	a static page that visualizes a network
Extend Session 1	INTERMEDIATE	more from your morning's cluster track
Recreate a paper	INTERMEDIATE	a result from a network science paper

Prompts, starter files, checklists: [hackathon/agentic/](https://hackathon.agentic/)

Track 1: Dashboard website

BEGINNER

A static site that visualizes a network, built by the agent

- Loads a graph, draws it (nodes sized by degree), shows stats + a degree histogram
- Plain HTML + JS (vis-network or D3), no build step. Starter `network.json` included
- Great on-ramp: the focus is the **workflow**, not hand-writing web code

▶ `hackathon/agentik/track-1-dashboard/`

Track 2: Extend your Session 1 work

INTERMEDIATE

Take this morning's cluster track further, with the agent

- Track A ▶ 2-D parameter sweep + heatmap
 - Track B ▶ sweep GNN architectures
 - Track C ▶ add an igrph comparison, or push the size ladder higher
 - Practice: a cluster-aware `CLAUDE.md`, `sbatch --test-only`, verify the real job output
- ▶ `hackathon/agentik/track-2-extend-session1/`

Track 3: Recreate a paper

INTERMEDIATE

Build this hierarchical network, then reproduce its result plots

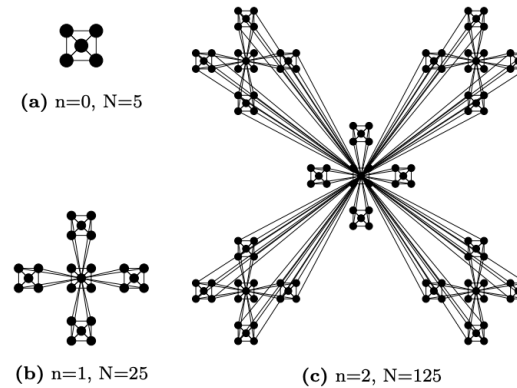
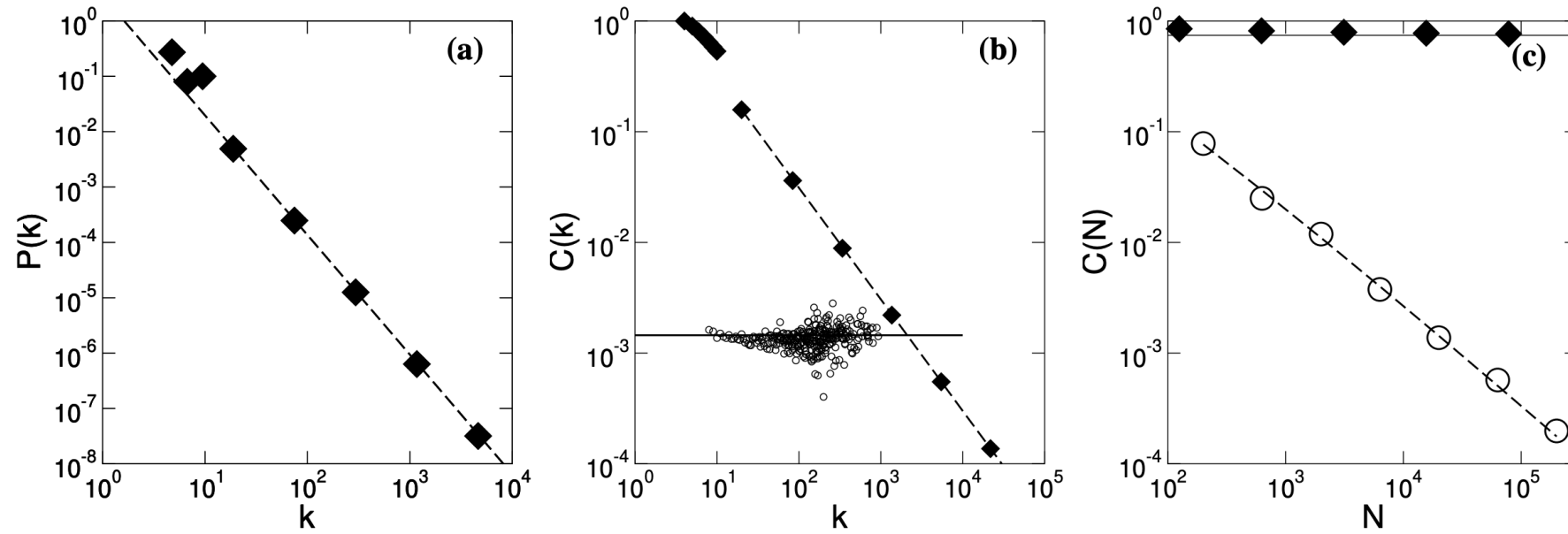


FIG. 1: The iterative construction leading to a hierarchical network. Starting from a fully connected cluster of five nodes shown in (a) (note that the diagonal nodes are also connected – links not visible), we create four identical replicas, connecting the peripheral nodes of each cluster to the central node of the original cluster, obtaining a network of $N = 25$ nodes (b). In the next step we create four replicas of the obtained cluster, and connect the peripheral nodes again, as shown in (c), to the central node of the original module, obtaining a $N = 125$ node network. This process can be continued indefinitely.

► hackathon/agentik/track-3-recreate-paper/ · arXiv [cond-mat/0206130](https://arxiv.org/abs/cond-mat/0206130)

The plots to reproduce



Your targets: **(a)** scale-free $P(k)$ · **(b)** $C(k) \sim 1/k$, the hierarchy fingerprint · **(c)** clustering stays constant as N grows

A good first prompt

Vague in, vague out. Give it a goal, context, and a done-condition:

```
Goal: a static index.html that loads network.json and draws the
      network with nodes sized by degree, plus a degree histogram.
Context: network.json has nodes[{id,degree}] and links[{source,target}].
Use vis-network from a CDN, no build step.
Done when: the page opens with no console errors and the node count
           shown on the page matches network.json.
Plan it first, then build in small steps.
```


Share-outs

60 seconds each

- **What I tried** (the goal)
- **What worked**
- **What broke**, and how you dealt with it
- **What I'd try next**

Takeaways

- Agents are a **force-multiplier, not autopilot**. You plan and verify
- Most of the value: a tiny `CLAUDE.md` + small, verifiable steps
- Best combo: **cluster + agent**. You did both today

Go use it on your actual research tomorrow.